

THE ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5 MODULE 3

PRACTICAL 1

UNIT CODE: 0612 451 07A

Unit: Network Design and Management

ELEMENT COVERED: Design computer network

Total Marks: 50 Marks

Duration: 2 Hours

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Instructions

- ✓ Read all instructions carefully before starting.
- ✓ Use Cisco Packet Tracer to complete the task.
- ✓ Save your file as:
- ✓ YourName_LogicalNetworkDesign.pkt
- ✓ Show all IP addressing and device configurations.
- ✓ Test connectivity before submitting.
- ✓ The Test consist of 4 pages

- ✓ Ensure you have all the pages

Materials

- ✓ A desktop computer installed with packet tracer software.
- ✓ 1 foolscap for documentation.

Scenario

ABC Training Institute has established a new computer laboratory with two departments:

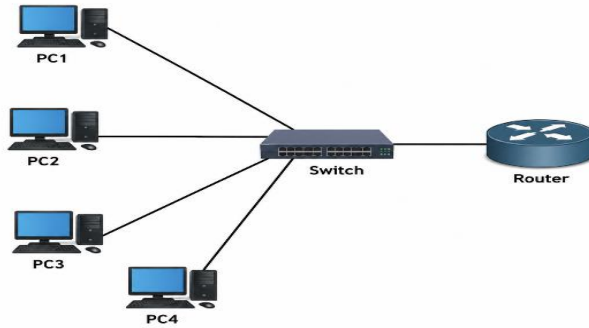
- Administration Department
- Training Department

The institute requires a logical network design that will allow communication between all computers through a router and switch.

Requirements

Design the network shown below:

Administration Department



Task 1: Create the Network Topology

Using Cisco Packet Tracer:

1. Add the following devices to the logical workspace:
 - 1 Router
 - 1 Switch
 - 4 PCs
2. Connect all devices using appropriate cables.
3. Arrange the devices neatly to form a logical network layout.

Task 2: Develop the Logical Network Design

Assign the following IP addresses:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.10.3	255.255.255.0	192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
PC3	192.168.10.4	255.255.255.0	192.168.10.1
PC4	192.168.10.5	255.255.255.0	192.168.10.1
Router interface G0/0	192.168.10.1	255.255.255.0	N/A

Configure all PCs with the correct IP address, subnet mask, and default gateway.

TASK 3: Adding training department logical design

Add 4 pcs and one switch on the other side using the same topology used on top

Connect the switch to the router using interface G0/2

Assign the following IP addresses to pcs in training deparment

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.2.2	255.255.255.0	192.168.10.1
PC2	192.168.2.3	255.255.255.0	192.168.10.1
PC3	192.168.2.4	255.255.255.0	192.168.10.1
PC4	192.168.2.5	255.255.255.0	192.168.10.1
Router interface G0/0	192.168.2.1	255.255.255.0	N/A

Task 4: Configure the Router

Configure the router interfaces connected to the switch

If your router uses FastEthernet interfaces, replace gigabitethernet0/0 with fastethernet0/0.

Task 5: Test Network Connectivity

Perform the following tests:

1. Ping from PC1 to PC2.
2. Ping from PC1 to PC4.
3. Ping from PC3 to the router.
4. Confirm that all devices can communicate successfully.

Record the results of each test.

Task 6: Save and Document the Network

Prepare a short report containing:

1. Title of the network.
2. Devices used.
3. IP addressing table.
4. Router configuration used.
5. Connectivity test performed.

Save the Packet Tracer file and submit it together with the report.